



Lab 03: DNS & DHCP Servers

DNS · DHCP · Google · Facebook Servers · Switch · PC0 · PC1 · PC2

Lab #03

Topic: Network Services

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Configure DHCP server to automatically assign IPs to PC0, PC1 and PC2.
- Configure DNS server to resolve google.lab and facebook.lab hostnames.
- Set up simulated Google and Facebook servers with static IPs.
- Verify DHCP lease acquisition and DNS resolution from all three PCs.
- Relate to Network+ objective 1.6: DNS and DHCP services.

2. Network Topology

Switch0 (2960) is the central device. DNS server connects on Fa0/5, DHCP on Fa0/1, Google server on Fa0/6, Facebook server on Fa0/7. PC0 on Fa0/2, PC1 on Fa0/3, PC2 on Fa0/4. All devices share the 192.168.1.0/24 subnet.

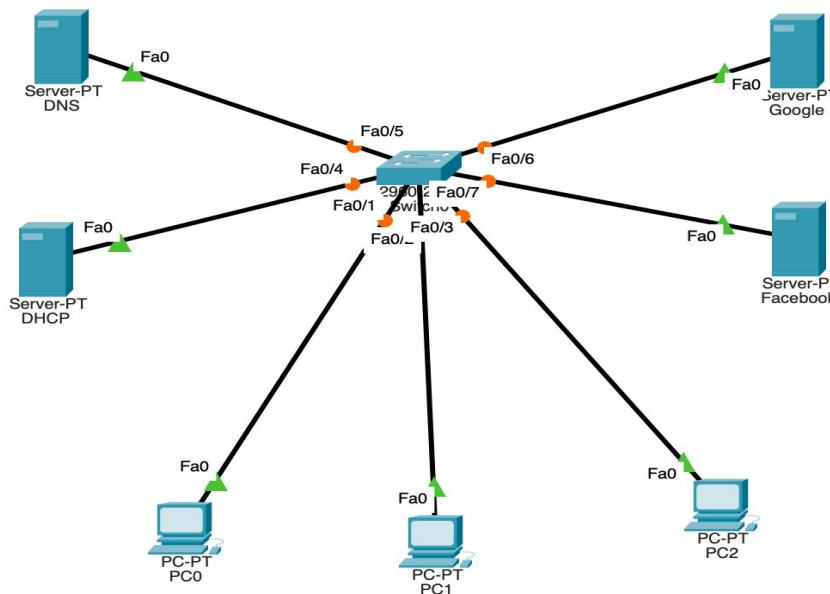


Figure 1 – Star topology: DNS, DHCP, Google, Facebook servers + PC0, PC1, PC2 on Switch0

Device	Role	Switch Port	IP Address	Notes
DNS	DNS Server	Fa0/5	192.168.1.2	Static — resolves hostnames
DHCP	DHCP Server	Fa0/1	192.168.1.3	Static — assigns IPs to clients



Google	Web Server	Fa0/6	192.168.1.10	Simulated google.lab
Facebook	Web Server	Fa0/7	192.168.1.20	Simulated facebook.lab
PC0	Client	Fa0/2	DHCP	Auto-assigned from pool
PC1	Client	Fa0/3	DHCP	Auto-assigned from pool
PC2	Client	Fa0/4	DHCP	Auto-assigned from pool

3. Configuration

3.1 DHCP Server

Pool Name	LAN_POOL
Default GW	192.168.1.1
DNS Server	192.168.1.2
Start IP	192.168.1.50
Subnet Mask	255.255.255.0
Max Users	50

3.2 DNS Records

Hostname	Type	IP Address
google.lab	A Record	192.168.1.10
facebook.lab	A Record	192.168.1.20
dhcp.lab	A Record	192.168.1.3

3.3 PCs – Enable DHCP

PC0/PC1/PC2 > Desktop > IP Configuration > select DHCP

Wait 2-3 seconds – IP, gateway and DNS auto-assigned

4. Verification

```
PC0>ipconfig
```

```
PC0>ping google.lab
```

```
PC0>ping facebook.lab
```

```
PC0>ping 192.168.1.2
```

5. Key Takeaways

- DHCP uses 4-step DORA: Discover, Offer, Request, Acknowledge.



- DNS A records map human-readable names to IPv4 addresses.
- DHCP option 6 (DNS) and option 3 (gateway) are distributed with the lease automatically.
- Simulating real services (Google, Facebook) builds practical DNS troubleshooting skills.
- Servers always get static IPs — DHCP is for client devices only in production networks.